ROBOTS MEET ARTS

EDUCATIONAL ROBOTICS AND CODING MEET ARTS AND HUMANITIES IN INCLUSIVE LEARNING ENVIRONMENTS

DOCUMENT TITLE

REPORT ON PROJECT'S BEST PRACTICES (PBP)

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1. Executive Summary

This report presents the key findings of the Robots Meet Arts project, which investigates how educational robotics and coding can support inclusive, interdisciplinary learning environments in primary education.

The project developed two complementary components:

- the Robot Meet Arts Teacher Training Programme designed to equip educators with the pedagogical and practical skills required to integrate robotics into classroom teaching
- the Robots Meet Arts Curriculum, which includes interdisciplinary robotics activities connecting coding with arts, languages, humanities, and social learning.

Both components were piloted in several European countries, including Greece, Spain, Belgium, and France. These diverse contexts allowed the project to examine how robotics can be applied in different educational settings and with learners who have varying levels of digital experience.

The analysis of the implementation activities resulted in the identification of best practices related to two main areas:

1. **Teacher training practices** that support educators in designing inclusive robotics lessons, understanding computational thinking, and integrating robotics across subjects.
2. **Curriculum implementation practices** that demonstrate effective ways to introduce robotics activities in classrooms through progressive learning pathways, collaborative learning structures, reflective practices, and interdisciplinary connections.

Across these practices, **several common pedagogical principles emerged**:

- robotics should be introduced progressively, starting with **accessible unplugged activities** and gradually moving toward **programming and robotics tools,**
- **inclusive design** should be considered during lesson planning to ensure participation of learners with diverse needs,
- **collaboration and peer learning** play a central role in supporting student engagement and problem-solving,
- **reflection and experimentation** help students understand computational concepts more deeply.

The findings demonstrate that educational robotics can serve as a powerful pedagogical tool for fostering digital competence, creativity, and critical thinking, while also supporting interdisciplinary learning and inclusive educational practices. The best practices presented in this report provide practical guidance for educators and policymakers seeking to integrate robotics into educational systems across Europe.

2. Introduction

Educational robotics is increasingly recognised as an effective pedagogical approach for developing key competences required in contemporary education, including computational thinking, creativity, collaboration, and problem-solving. When integrated thoughtfully into teaching practices, robotics can support interdisciplinary learning that connects science, technology, arts, and humanities while fostering active and inclusive student participation.

The Robots Meet Arts project explores how educational robotics and coding can enrich learning environments by bridging technology with arts and humanities subjects in primary education. The project focuses particularly on inclusive educational practices, aiming to demonstrate how robotics can support diverse learners while promoting digital competences and creative thinking.

To achieve this objective, the project developed and implemented both the **Robots Meet Arts Teacher Training Programme** and the **Robots Meet Arts Curriculum**, which were piloted across several European contexts. Implementation activities took place in **Greece, Spain, Belgium, and France**, where the curriculum was tested during an educational event focused on robotics, coding, and artificial intelligence. These varied contexts provided valuable opportunities to test robotics activities in different institutional, pedagogical, and socio-cultural environments.

Teachers participating in the project often had limited prior experience with educational robotics, allowing the project to examine how professional development and structured learning materials can build teachers' confidence and support the effective integration of robotics into classroom practice.

Based on these implementation experiences, the project identified a **series of best practices related to both teacher training and curriculum implementation**. These practices highlight effective pedagogical strategies for introducing robotics, supporting inclusive participation, and connecting computational thinking with meaningful learning experiences across different subjects.

This report presents these findings and aims to provide practical guidance for educators, schools, and policymakers seeking to integrate educational robotics into teaching and learning across European educational contexts.

3. Overview of the Implementation Contexts

The identification of best practices within the Robots Meet Arts project is grounded in the implementation experiences carried out across different educational contexts in Europe. The project partners piloted the Teacher Training Programme and the Robots Meet Arts Curriculum in a variety of institutional, pedagogical, and socio-cultural environments. These contexts provided valuable insights into how robotics and computational thinking can be integrated into arts, humanities, and interdisciplinary learning in primary education.

Implementation activities took place in **Greece, Spain, Belgium, and France**, where partners held the Robots Meet Arts teacher training programme and piloted the Robots Meet Arts curriculum with students in real classroom settings. The participating schools and training environments reflected **diverse educational realities, including urban and rural schools, socially diverse communities, and classrooms with varying levels of experience with digital technologies and educational robotics**. In most cases, teachers had **limited prior experience with robotics**, which allowed the project to test the effectiveness of the training programme in building **confidence, pedagogical competence, and practical classroom applications.** In addition to these planned implementation contexts, an **additional piloting activity was carried out at CERN,** where the project curriculum was tested during a full-day educational event dedicated to robotics, coding, and artificial intelligence. This setting offered the opportunity to observe the activities with a broader range of learners, including students from different school systems and age groups.

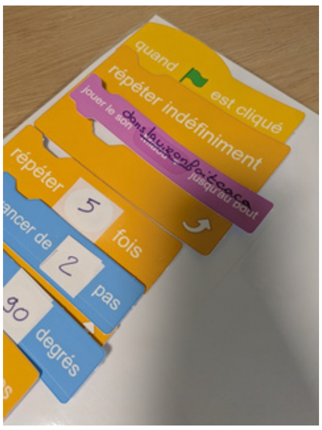
Across the five contexts, the project reached teachers and students with different levels of prior knowledge, technological access, and educational needs. This diversity provided a rich evidence base for identifying successful approaches, transferable strategies, and effective pedagogical practices. The **following sections present an overview of each implementation context**, which serves as the foundation for the identification and analysis of the project's best practices.

1. Overview of the Implementation Context – France

In France, the Robots Meet Arts implementation took place in two complementary settings: initial in-service teacher training within a teacher education institute and curriculum piloting in an out-of-school support programme aimed at learners from disadvantaged contexts.

The teacher training was implemented at **INSPÉ Créteil (Paris)** through two cohorts. **The first cohort took place on 03/03/2025, 07/03/2025, 20/05/2025, and 22/05/2025, reaching 22 in-service primary teachers across eight thematic sessions delivered over six half-days**. A second cohort followed in **January 2026 with 17 teachers, covering six sessions over four half-days**, with the remaining modules planned for May 2026. Participants were generalist primary teachers responsible for all subjects, including humanities, and most reported limited prior familiarity with computational thinking, coding, or educational robotics. The training design combined conceptual input with active pedagogy and "unplugged-to-plugged" progression, including activities such as embodied programming, narrative-based computational thinking for humanities, networking simulations, collaborative design tasks, and explicit reflection on inclusion. Trainers reported strong engagement and collaboration, and very high satisfaction and recommendation rates, while noting that making interdisciplinary links explicit required additional facilitation and that confidence to implement independently was higher after completing the full training cycle.

Curriculum piloting was carried out in **La Rochelle within the CLAS (Contrat Local d'Accompagnement à la Scolarité) programme**, which supports young people outside school hours, particularly in priority neighbourhoods. The piloting involved **three sessions on 12/03/2025, 16/04/2025, and 07/05/2025, with one facilitator/teacher and nine students from the La Pallice area**. The group was mixed-age, ranging from CM2 (10–11 years old) to 4ème (13–14 years old), and included a high proportion of learners requiring adapted support, with around half presenting ADHD or ASD. The piloted activities combined unplugged and digital approaches and were implemented through a strongly multimodal methodology that supported engagement, peer learning, and inclusion. Feedback from both teachers and students indicated high motivation and enjoyment, strong perceived learning, and good usability of materials, while challenges related to mixed-age dynamics, attention regulation, and time constraints for creative projects highlighted the importance of careful pacing and scaffolding in similar contexts.



In addition to the implementation in La Rochelle of the project, a curriculum piloting activity was carried out with teachers from cross-border schools **at CERN**. This pilot provided an opportunity to test a Robots Meet Arts activity with a wider age range and highly diverse student profiles, extending the project's evidence base beyond the original partner-country plan.

The piloting took place on **14 November 2025 as part of a full-day CERN initiative inviting teachers to bring their classes to discover activities in robotics, coding, and artificial intelligence.** In total, **four teachers and fifty students participated in four sessions of 1h30 each**, welcoming four different classes spanning **8 to 16 years old**. Two groups came from either French cross-boarder schools or international schools. Gender distribution across the participating groups was reported as well balanced.

The piloted activity was the Robots Meet Arts lesson "**Born Yesterday – Bio-inspired Learning Processes and Robotics**." The implementation combined an unplugged trial-and-error learning game with robot programming workshops, allowing students to translate a self-built learning model into executable instructions. Students rotated through three different programming environments (Cubetto, Sphero Indi, and micro:bit-based robots) and concluded by comparing their learning process to an AI approach using an external tool.

Satisfaction feedback from both teachers and students was consistently very high across all criteria, highlighting strong motivation, effective collaboration, clear facilitation, sufficient timing, and successful participation for learners with very different starting points. Key challenges related mainly to managing the wide range of prior experience and the need for clearer pre-event communication about class suitability, while the most notable success was the activity's adaptability and universal accessibility, enabling both beginners and experienced students to find meaningful challenge and learning.



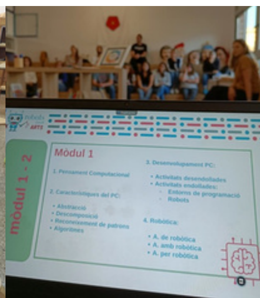
2. Overview of the Implementation Context – Spain

In Spain, the Robots Meet Arts implementation combined a hybrid teacher training model with curriculum piloting in a public primary school, in a context where inclusion, transversal learning, and the integration of digital tools are established priorities. Across both components, participants showed strong motivation to innovate, while prior experience with robotics and programming varied from beginner to occasional classroom use.

The teacher training was delivered as a 30-hour course, with 22 hours online and 8 hours face-to-face, following a structure that alternated conceptual work with practice-oriented sessions. The training was attended by **33 in-service early childhood and primary education teachers** from the areas of **Tarragona & Reus in Catalonia**. Participants had diverse professional trajectories and solid teaching experience. Online modules focused on the foundations of educational robotics, programming, and computational thinking, while face-to-face meetings emphasised experimentation with materials and the design of classroom-oriented activities. A key feature of the Spanish implementation was the systematic use of entry and exit tickets during the face-to-face sessions, supporting formative assessment, reflection, and continuous adaptation of the training to participants' needs. Trainers reported very high engagement, interaction, timing, and collaboration, while noting challenges related to scheduling face-to-face attendance and addressing the heterogeneous prior knowledge of participants through differentiated support.

Curriculum piloting took place in **Reus at Escola Marià Fortuny on 28–30 January 2026**, involving six teachers and sixty students aged 8–11. The school is located in an urban area of Catalonia and has previous experience with interdisciplinary projects integrating robotics, arts, and digital storytelling. Students generally had basic to intermediate familiarity with digital technologies, while the school context included learners from low to medium socioeconomic backgrounds and some situations of social vulnerability, as well as diverse learning needs and multilingual profiles. Across three piloting sessions, activities integrating robotics with arts, languages, and history were implemented through collaborative structures and differentiated materials. Teacher and student feedback indicated high motivation and enjoyment, strong perceived learning, and very positive inclusion outcomes, while logistical issues such as device management and technical preparation time were identified as areas for improvement.





Co-funded by the European Union

3. Overview of the Implementation Context – Greece

In Greece, the Robots Meet Arts implementation took place in a primary school setting and included two linked components: a teacher training programme and the piloting of selected curriculum activities with students. The context is characterised by strong diversity and inclusion needs, as well as educators who are motivated to innovate but, in many cases, have limited prior experience with educational robotics.

The Teacher Training Programme was delivered on 3–4 September 2025 at the **Dimotiko Scholeio of Platy**, with 33 participants from Imathia region attending 8 sessions. Participants were mainly primary teachers, with additional representation from special education and school leadership. Pre-training data showed low prior experience with robotics and coding, indicating a clear need for capacity building. The training combined introductions to computational thinking and the project curriculum with hands-on exploration of tools such as Bee-Bot, Robot Mouse, Indi, Scratch Jr., LEGO Spike Essential, and Vinci Robot, and included collaborative group work to translate learning into classroom practice. Overall evaluation feedback indicated very high engagement and satisfaction, while also suggesting that some teachers would benefit from additional practice time to build confidence.

The Curriculum Piloting took place on 2 and 4 December 2025 through 3 sessions, involving 6 teachers and 52 students aged 9–11 at the **Dimotiko Scholeio of Platy**. Three activities were piloted (Act Green Think Fair, Dilemmas with Values, and Treasure Hunters), using tools including Tale-Bot, Scratch, and LEGO Spike Essential. The student cohort was highly diverse, including Roma students, multilingual learners, and pupils with special educational needs, requiring strong scaffolding and differentiated support. Co-teaching (general and special education) and peer mentoring supported inclusive participation. Feedback from both teachers and students showed high motivation and enjoyment, strong inclusion outcomes, and good usability of materials, while time constraints and language-related barriers emerged as key challenges.

4. Overview of the Implementation Context – Belgium

The activities took place within the **Flemish education system, where schools are increasingly expected to integrate computational thinking and ICT across subject areas**. The implementation therefore focused on supporting teachers in developing practical strategies to incorporate robotics and programming into humanities, arts, and other subjects.

The Teacher Training Programme was organised in **Heverlee (Belgium) and involved 30 participants, mainly primary school teachers, along with a small number of special education teachers, STEAM or ICT coordinators, representatives of community organisations, and students in a programming associate degree.** The training was delivered through two intensive full-day sessions (7 and 12 November 2025) followed by two online follow-up sessions (10 and 12 December 2025), covering eight modules. Prior to the training, participants were introduced to the Moodle platform and completed a survey on their background and experience with robotics. The results showed that although participants were experienced educators, many had limited prior experience with educational robotics, highlighting the need for targeted professional development. The training combined theoretical input with hands-on exploration of a wide range of tools, including Makey Makey, Tale-Bot, Indi Sphero, Edison, Photon, Micro:bit, Scratch extensions, and Cubetto.

The Curriculum Piloting took place on **25 November 2025 at De Ark School in Kessel-Lo (Leuven) and involved three teachers and three classes of students aged 10–12 (approximately 66 students in total)**. The participating classes represented diverse learning environments, including pupils with different language backgrounds, varied ability levels, and some students following adapted learning programmes. Three curriculum activities were implemented: Human Robot Prepares Breakfast, Making Faces with OctoStudio, and Dance Mode. These activities combined unplugged programming, creative coding, and embodied learning approaches to introduce students to sequencing, algorithms, collaboration, and problem-solving. Feedback from teachers and students indicated strong motivation and interest in robotics activities, particularly because they offered new and engaging learning experiences. At the same time, the implementation highlighted some challenges related to language diversity, classroom management during collaborative tasks, and technical restrictions on school devices.




4. Best Practices from the Implementation of the Robots Meet Arts Teacher Training Programme

Teacher professional development plays a central role in the successful integration of robotics and computational thinking into school education. Within the Robots Meet Arts project, teacher training activities were implemented in several European contexts (in France, Belgium, Spain and Greece) bringing together educators with diverse levels of experience in digital technologies, robotics, and innovative pedagogies. The training programmes aimed not only to introduce teachers to educational robotics but also to support them in developing the confidence and pedagogical strategies needed to integrate these tools meaningfully into their teaching practice.

Across the different pilot contexts, partners experimented with a **variety of training approaches designed to address common barriers faced by teachers**, such as limited experience with coding, technological anxiety, or uncertainty about how robotics can be connected to non-STEM subjects. The training programmes therefore emphasised hands-on experimentation, collaborative learning, inclusive pedagogical design, and interdisciplinary applications linking robotics with arts, humanities, and language learning. Through the analysis of the training activities and feedback collected from participants, **twelve best practices were identified**. These practices reflect **effective strategies for designing teacher training programmes that promote engagement, accessibility, and long-term classroom implementation of educational robotics and computational thinking**.

The **best practices presented** in the following section highlight different aspects of successful professional development, including experiential learning, inclusive lesson design, collaborative training formats, formative assessment techniques, and low-tech approaches that make computational thinking accessible even in contexts with limited technological resources. Together, they provide **practical guidance for teacher trainers, and policy makers seeking to strengthen teachers' digital and pedagogical competences across Europe**.

Overview of Best Practices in Teacher Training

Best Practice Title	Country	Short Description
• A Progressive Approach to Teaching Computational Thinking: From Unplugged to Digital	France	• Computational thinking concepts are introduced through physical unplugged activities before transitioning to digital programming tools.
• AI-Enhanced Activities for Accelerated Creation and Inclusive Learning	France	• AI tools support teachers in generating creative activities and adapting learning materials to diverse classroom needs.
• Low-Tech Simulation of Digital Systems in Teacher Training	France	• Everyday objects such as balls, cards, and printed materials are used to simulate algorithms and digital communication processes.
• Hands-on Teacher Training through a “Robot Market” Discovery Approach	Belgium	• Teachers explore and test a variety of educational robots in a discovery-based training environment that encourages experimentation.
• A Library-Based Robot Lending System for Sustainable Access to Educational Robotics	Belgium	• A lending system provides schools with free access to robotics equipment, reducing financial barriers to adoption.
• Introducing Computational Thinking through Unplugged Experiential Activities	Belgium	• Teachers experience programming concepts through playful unplugged activities before working with digital tools.
• Hybrid Teacher Training Combining Online Preparation and Hands-On Collaborative Practice	Spain	• Training combines self-paced online modules with face-to-face sessions focused on experimentation and collaboration.
• Teaching Computational Thinking through a Structured Theory-to-Practice Approach	Spain	• Teachers first explore key theoretical concepts and then apply them through practical robotics and programming activities.
• Entry and Exit Tickets for Continuous Reflection in Teacher Training	Spain	• Short reflection activities before and after sessions help trainers monitor learning progress and adapt the training.
• Collaborative Design of Robotics Lesson Plans	Greece	• Teachers work in groups to design robotics-based learning activities that can be directly implemented in classrooms.
• Simulation-Based Teacher Training for Inclusive Robotics Education	Greece	• Teachers experience robotics lessons as learners through simulated classroom activities before designing their own.
• Inclusive Lesson Design with Educational Robotics	Greece	• Teachers design robotics activities that intentionally address the needs of diverse learners and promote inclusive participation.

1. A Progressive Approach to Teaching Computational Thinking: From Unplugged to Digital

General Information

Country: France
Partner Organisation: L.A.B (Laboratoire d'Aix-périmentation et de Bidouille)
Type of Best Practice: Teacher Training

Description of the Practice

Throughout the ten-half-day training programme (March 2025 to January 2026) delivered to 39 primary school teachers at INSPÉ Créteil, the training followed a deliberate **unplugged-to-plugged progression**. Each computational thinking concept was first introduced through a **physical, hands-on activity before any digital tool was involved**. In "Dance Your Block!", teachers coded a choreography with printed Scratch blocks; in "Pass the Parcel", they simulated packet routing by passing balls; in "Narrative Maze", they built branching stories with cards and dice. The consistent pattern — feel the concept first, then formalise it digitally — was applied across all sessions.

Why is this a Best Practice?

This approach removed the **initial barrier of technical anxiety that generalist teachers experience with coding or robotics**. The unplugged phase built intuitive understanding and confidence before digital formalisations. Teachers could relate abstract concepts to concrete, embodied experiences, making the transition to digital tools feel natural. Unplugged activities also made interdisciplinary links with humanities more visible, as they inherently involve storytelling, collaboration, and creative expression.

Impact

Survey results showed 100% of participants found activities engaging (77.27% strongly agree, 22.73% agree). Teachers highlighted that **starting unplugged helped them "feel" concepts more easily**. Trainers observed the **progression significantly reduced resistance to computational thinking**, and the **turnkey nature of unplugged activities** was particularly appreciated for direct classroom reuse.

Transferability

The progression requires **no digital equipment for its initial phase** — only printed cards, balls, dice, or body movement — making it **accessible in low-resource settings**. The approach is **subject-agnostic** and adaptable to any age group. It is particularly suitable where teachers have **limited prior experience with coding**, as gradual confidence-building supports acculturation. Partners can design their own unplugged activities using the Robots Meet Arts activity bank as a starting point.

2. AI-Enhanced Activities for Accelerated Creation and Inclusive Learning

General Information

Country: France
Partner Organisation: L.A.B (Laboratoire d'Aix-périmentation et de Bidouille)
Type of Best Practice: Teacher Training

Description of the Practice

During the 2 specific sessions of the L.A.B training programme, we demonstrated how **generative AI tools could be integrated into computational thinking activities** to both **accelerate the creative process** and **support inclusive participation**. Taking the "Narrative Maze" activity as a key example, teachers first collaboratively defined a narrative framework and character sheets. They then used AI prompts to generate a complete branching story and a printable maze layout, drastically reducing preparation time. Teachers explored how AI could serve as a creative scaffolding tool — not replacing the pedagogical design, but accelerating content production and enabling real-time adaptation to diverse learner needs.

Why is this a Best Practice?

This practice showed that **AI**, properly used, can unlock **creative exploration** while not replacing pedagogical design. Teachers who felt unsure about inventing a narrative or designing a maze from scratch could **generate a first draft instantly and reshape it**. This lowered the **entry barrier for less confident creators** and opened up unexpected directions. It also proved a **powerful inclusivity lever**: generating simplified texts, visual aids for non-native speakers, or adapted story paths became as simple as adjusting a prompt, making differentiation a creative act rather than an extra burden.

Impact

Teachers described the AI integration as a shift in their creative process: from blank-page anxiety to rapid prototyping. The group spontaneously used AI-generated content as a base for the lesson plans they designed. The approach made differentiation feel accessible and creative rather than technical.

Transferability

This practice requires only access to a free AI chatbot (ChatGPT, Claude, Mistral). The transferable method is simple: start from a structured template, then prompt AI to generate or vary content. It works with any activity and any language and is especially powerful for producing **inclusive materials adapted to local contexts** including language barriers, special needs, or cultural diversity.

3. Low-Tech Simulation of Digital Systems in Teacher Training

General Information

Country: France
Partner Organisation: L.A.B (Laboratoire d'Aix-périmentation et de Bidouille)
Type of Best Practice: Teacher Training

Description of the Practice

During the teacher training programme delivered at INSPÉ Créteil, several activities were designed to **simulate digital processes using everyday objects such as balls, cards, and printed materials**. Instead of immediately introducing computers or programming tools, teachers first explored how digital systems work through simple physical simulations.
For example, in the activity "Pass the Parcel", participants simulated how information travels through a network by passing objects between participants according to predefined rules. Through this activity, teachers experienced how messages can be transmitted, delayed, or redirected within a communication network. Other activities used cards or printed materials to represent algorithms and programming logic. **These simulations allowed teachers to explore computational concepts such as data transmission, sequencing, and algorithmic processes in an accessible and interactive way before moving to digital tools.**

Why is this a Best Practice?

This approach also reduces the technological barrier often associated with coding and robotics. Teachers discover that computational thinking can be introduced through simple and playful activities, making it easier to integrate these concepts into their own teaching practice.

Impact

Teachers reported that these activities helped them better **understand how digital systems operate and gave them practical ideas for introducing computational thinking in their classrooms**. The approach demonstrated that complex technological concepts can be explained through simple classroom activities that encourage participation, collaboration, and experimentation.

Transferability

This practice is highly transferable because it requires **minimal resources and no digital equipment**. Activities can be implemented using common classroom materials such as cards, balls, or printed instructions. As a result, the approach can easily be adapted to different educational contexts, age groups, and subjects, making it particularly suitable for schools with limited technological resources.

4. Hands-on Teacher Training through a “Robot Market” Discovery Approach

General Information

Country: Belgium
Partner Organisation: UCLL
Type of Best Practice: Teacher Training

Description of the Practice

During the teacher training sessions, participants explored several educational robots through a **“robot market” format**. **A wide range of tools—including Photon, Micro:bit, Edison, TaleBot, and Sphero Indi—were made available for hands-on experimentation**. **Teachers rotated between stations, testing the robots independently, discussing their experiences with peers, and asking questions to trainers.** Teachers also worked with the Robots Meet Arts lesson plans for the different subject areas. This structure created an open and flexible learning environment where teachers could discover different programming environments and pedagogical possibilities.

Why is this a Best Practice?

The **“robot market” format** promotes discovery-based learning and active experimentation during teacher training. Instead of receiving only demonstrations, teachers are able to **explore robots and lesson plans independently and at their own pace**, which encourages **curiosity** and **confidence** in using new technologies. It allows participants to compare different tools, reflect on how they might apply them in their classroom practice and supports differentiated learning during training, as participants with different levels of experience can explore simple or more advanced tools.

Impact

Teachers reported that hands-on experimentation helped them better understand the pedagogical potential of different robots and how they could be integrated into classroom activities. The opportunity helped participants reflect critically on which technologies might best suit their teaching context. The interactive format i**ncreased engagement** and promoted **peer exchange**, allowing teachers to share ideas and practical experiences throughout the training.

Transferability

This approach can **easily be replicated** in other teacher training contexts. It **requires a selection of educational robots, a flexible workshop space, and facilitators who can support participants during experimentation.** The “robot market” format is adaptable to different training durations and can be implemented in teacher education institutions, schools, or professional development programmes.

5. A Library-Based Robot Lending System for Sustainable Access to Educational Robotics

General Information

Country: Belgium
Partner Organisation: UCLL
Type of Best Practice: Teacher Training

Description of the Practice

Through the Robots Meet Arts project, UCLL acquired around fifty educational robots—including Photon, Micro:bit, Edison, TaleBot, and Sphero Indi—which are now available through the UCLL library. **These robots can be borrowed free of charge by teachers, students, and other users.** The robots were introduced during the teacher training sessions and **remain accessible after the project's completion.** The **lending system** allows educators to experiment with different tools in their own teaching environment before deciding whether to invest in them.

Why is this a Best Practice?

Providing **free access to educational robots removes financial barriers that often prevent schools from experimenting with innovative technologies**. Many schools lack the budget to purchase multiple robots for classroom use, making a **lending system an inclusive solution**. Allowing teachers to test different robots before making purchasing decisions helps them choose intentionally rather than relying on assumptions. This approach **reduces the risk of costly misinvestments** and increases the long-term adoption of meaningful digital tools.

Impact

Through this initiative, teachers **gain equal access to a wide range of educational robots** regardless of their school's financial situation. Schools can avoid unnecessary purchases and ensure that robots are used effectively rather than remaining unused in storage. The **sustainability of the lending system ensures continued access to robotics tools and ongoing impact beyond the duration of the project.**

Transferability

This **model can be replicated by universities**, **teacher training institutions, libraries, or educational networks that have the capacity to manage shared equipment**. By centralising resources and providing borrowing systems, institutions can lower the financial barrier for schools and support teachers in experimenting with robotics before making purchasing decisions. The model is particularly suitable for regional teacher training centres and innovation hubs.

6. Introducing Computational Thinking through Unplugged Experiential Activities

General Information

Country: Belgium
Partner Organisation: UCLL
Type of Best Practice: Teacher Training

Description of the Practice

During the teacher training sessions on 7 and 12 November in Belgium, theoretical explanations were combined with hands-on activities. Participants tested an **unplugged lesson using printed Scratch blocks**. **Programming icons were secretly placed under two participants' chairs, and those who found them were invited to create a short "program" for a dance sequence using the blocks**. The whole group then performed the programmed movements together, translating code into physical actions and experiencing computational thinking in practice.

Why is this a Best Practice?

This practice helps teachers experience **computational thinking in an accessible and engaging way before using digital tools**. By **starting with an unplugged activity, technological barriers and fears associated with coding are reduced**. Teachers learn programming concepts such as sequencing and algorithms through a playful and collaborative activity, experiencing the learning process from a student's perspective.

Impact

The participants described the activity as a true eye-opener, giving them **confidence** to begin teaching computational and block-based coding in their own classrooms. For many teachers, this was the critical first step toward **understanding how coding can be introduced in an accessible way**. Several teachers indicated that they now plan to **start with an unplugged activity before moving on to a digital (plugged) exercise**. As a result, this practice not only strengthened their skills but also positively influenced their future instructional decisions.

Transferability

This practice can be **replicated across EU contexts because it relies on universal, low-threshold teaching principles rather than country-specific curricula**. The unplugged format makes computational thinking accessible to teachers regardless of their prior experience with coding. Its playful and inclusive structure fits well within diverse educational systems and classroom cultures across Europe.

7. Hybrid Teacher Training Combining Online Preparation and Hands-On Collaborative Practice

General Information

Country: Spain
Partner Organisation: URV
Type of Best Practice: Teacher Training

Description of the Practice

The teacher training programme followed a **hybrid format, combining online modules with face-to-face sessions focused on practical experimentation**. The online component allowed teachers to study theoretical foundations **at their own pace**, while the in-person sessions provided opportunities to **test robotics activities and collaboratively design classroom applications.**

Why is this a Best Practice?

The hybrid structure allowed teachers to prepare conceptually online and apply knowledge immediately during hands-on sessions. This approach helped **reduce cognitive overload and supported deeper understanding**. Small-group work promoted collaboration and peer learning, while progressive challenges increased motivation and confidence in using robotics in teaching.

Impact

The training strengthened **teachers' confidence and practical skills in educational robotics**. Participants demonstrated the ability to integrate theory and practice when designing lesson plans and reported feeling more prepared to apply robotics activities in their classrooms. Reflective discussions also helped teachers anticipate potential classroom challenges.

Transferability

The hybrid model can be easily **adapted to different educational systems and languages**. To replicate the practice, it is important to combine self-paced online preparation with practical face-to-face sessions, organise activities in small collaborative groups, and provide clear guidance and progressively challenging tasks.

8. Teaching Computational Thinking through a Structured Theory-to-Practice Approach

General Information

Country: Spain
Partner Organisation: URV
Type of Best Practice: Teacher Training

Description of the Practice

The teacher training programme introduced participants to the **core concepts of Computational Thinking (CT) and then progressively connected these concepts to classroom practice**. Teachers first explored the key CT skills and their **relevance for education, and then examined how these skills could be applied across different educational levels and subject areas**. The training combined theoretical explanations with practical examples using robotics and programming activities.

Why is this a Best Practice?

The approach was effective because it provided teachers with a **clear conceptual foundation before introducing practical applications**. This progression helped teachers understand **not only how to use robotics and programming tools but also why they support learning**. Collaborative activities and practical examples further supported the transfer of knowledge to classroom practice.

Impact

The training increased teachers' confidence in teaching computational thinking and strengthened their ability to integrate robotics and programming into lesson design. Teachers reported a better understanding of how computational thinking can be applied across subjects.

Transferability

This approach can be easily replicated in other teacher training programmes. It requires **structured sessions that introduce the core concepts of computational thinking** before moving to practical classroom applications. Combining theoretical explanations with hands-on activities helps teachers understand both the pedagogical value and the practical implementation of computational thinking.

9. Entry and Exit Tickets for Continuous Reflection in Teacher Training

General Information

Country: Spain
Partner Organisation: URV
Type of Best Practice: Teacher Training

Description of the Practice

During the face-to-face sessions of the teacher training programme, **entry and exit ticket** activities were systematically used to support **reflection and formative assessment**. At the beginning of each session, participants completed a short entry ticket where they briefly shared their prior knowledge, expectations, or questions related to the topic of the session. This helped trainers identify teachers' starting points and adapt the learning activities accordingly. At the end of each session, participants completed an exit ticket, reflecting on what they had learned, difficulties they encountered, and how they could apply the new knowledge in their teaching practice. These short reflections encouraged teachers to consolidate their understanding and provided trainers with valuable feedback about the effectiveness of the session.

Why is this a Best Practice?

Entry and exit tickets are an effective strategy for formative assessment in teacher training. They allow trainers to **identify participants' prior knowledge and misconceptions at the beginning of a session, making it possible to adjust the pace and focus of the training**. At the same time, exit tickets encourage teachers to **reflect** on their learning **process and connect new concepts with their classroom practice**. This approach strengthens engagement and promotes reflective practice, which is essential for meaningful professional development. The activity requires minimal time and resources while providing valuable insights into participants' learning progress.

Impact

Trainers were able to adapt activities to teachers' needs, while participants reported that the reflective prompts helped them clarify key concepts and think about how to apply robotics and computational thinking in their classrooms.

Transferability

This practice can easily be implemented in other teacher training programmes across Europe. Entry and exit tickets require minimal preparation and can be applied in both online and face-to-face training contexts. They can be adapted to different topics, training formats, and participant profiles, making them a **flexible tool for supporting reflection and formative assessment in professional development activities.**

Co-funded by the European Union

10. Collaborative Design of Robotics Lesson Plans

General Information

Country: Greece
Partner Organisation: STIM
Type of Best Practice: Teacher Training

Description of the Practice

Participants worked in **small groups to design robotics-based learning scenarios** tailored to specific subjects, age groups, and inclusion contexts. Teachers used a **lesson plan template and a variety of robotics equipment**, including Bee-Bot, Robot Mouse, Indi, LEGO Spike Essential, Scratch, and Vinci Robot to develop interdisciplinary lesson plans integrating robotics into different subjects. The activity was fully hands-on and collaborative. Each group presented its lesson plan to the plenary, followed by reflection and feedback.

Why is this a Best Practice?

This practice was particularly effective because it **moved beyond theoretical training** and required teachers to actively apply their learning by **collaboratively** designing real classroom scenarios. Instead of passively exploring robotics tools, participants translated their experience into **concrete, curriculum-aligned lesson plans ready for implementation**.

Impact

The activity had a strong impact on teachers' professional development. It increased their **confidence** in using robotics tools, strengthened their **ability to integrate computational** thinking across disciplines, and enhanced their skills in **designing inclusive and differentiated lesson plans**. Despite the initially low levels of familiarity with robotics, participants demonstrated substantial progress in translating robotics activities into practical classroom applications. The experience also fostered a sense of professional community, as teachers collaborated, exchanged ideas, and reflected collectively on inclusive pedagogical strategies.

Transferability

This best practice can be replicated in other European educational contexts. It requires **sufficient time for collaborative design, access to at least a small range of robotics tools, and a clear lesson planning template**. It is particularly suitable for contexts where teachers have limited prior experience with robotics, as the collaborative and guided nature of the activity supports gradual confidence-building. Its strong emphasis on interdisciplinary integration and inclusive design makes it adaptable to different national curricula and diverse classroom realities.

11. Simulation-Based Teacher Training for Inclusive Robotics Education

General Information

Country: Greece
Partner Organisation: Dimotiko Scholeio Plateos Imathias
Type of Best Practice: Teacher Training

Description of the Practice

During the two-day Teacher Training Programme (3–4 September 2025), teachers participated in **simulation-based activities** using Bee-Bot and Tale-Bot robots. Instead of only observing demonstrations, teachers first experienced robotics activities as learners before designing their own lesson plans. Through hands-on tasks such as storytelling, sequencing, and vocabulary activities with the robots, teachers explored how robotics can support literacy learning. At the same time, they reflected on how these activities could be adapted for diverse learners, including students with learning disabilities, autism spectrum conditions, multilingual backgrounds, and girls who may need additional encouragement in STEM-related activities.

Why is this a Best Practice?

Simulation-based learning allowed teachers to **experience robotics activities from the student perspective**. This approach helped **reduce initial technological anxiety** and supported a **deeper understanding** of how robotics can be integrated into inclusive classroom practice. By designing lessons immediately after the simulation, teachers were able to translate their experience directly into pedagogical planning.

Impact

Teachers reported **increased confidence** in using robotics tools and designing inclusive learning activities. The lesson plans produced during the training demonstrated thoughtful differentiation and creative integration of robotics into literacy learning.

Transferability

This practice can **easily be replicated in other teacher training contexts**. It requires accessible robots such as Bee-Bot or Tale-Bot and a training structure where teachers first experience activities through simulation before designing their own lessons. The approach can be adapted to **different subjects, age groups, and training formats**.

12. Inclusive Lesson Design with Educational Robotics

General Information

Country: Greece
Partner Organisation: Dimotiko Scholeio Plateos Imathia
Type of Best Practice: Teacher Training

Description of the Practice

During the two-day Teacher Training Programme (3–4 September 2025), teachers explored how educational robots such as Bee-Bot and Tale-Bot can be used to design learning activities that promote inclusion. Participants were encouraged to create lesson plans that ensured the meaningful participation of **all students, including learners with learning disabilities, students on the autism spectrum, children from multilingual or culturally diverse backgrounds, and girls** who may need additional encouragement to engage with STEM-related activities. Teachers worked through hands-on activities involving storytelling, sequencing, and language development while programming the robots. Through these activities, they explored how **robotics tasks could be adapted through scaffolding, differentiated instructions, visual supports, and collaborative learning structures**. The training emphasised that **inclusive strategie**s should be considered during the initial planning stage of the lesson, rather than being added afterwards.

Why is this a Best Practice?

This approach helps educators understand how **robotics can support accessibility, collaboration, and differentiated learning**. By embedding inclusion in the lesson design process, teachers developed a more holistic understanding of how robotics activities can engage all students and create equitable learning opportunities.

Impact

Teachers reported **increased confidence in designing robotics activities that are accessible to diverse learners**. The lesson plans produced during the training reflected thoughtful differentiation strategies and creative uses of Bee-Bot and Tale-Bot across different learning needs. Participants demonstrated a stronger awareness of how robotics can support inclusive and collaborative classroom environments.

Transferability

This practice can be easily implemented in other teacher training contexts. It requires accessible programmable robots and a training design that encourages teachers to consider inclusion during the planning of robotics activities. Because the approach focuses on pedagogical principles rather than specific curriculum requirements, it can be adapted to different subjects, school systems, and educational contexts across Europe.

5. Best Practices from the Implementation of the Robots Meet Arts Curriculum

Educational robotics can play a transformative role in school curricula by supporting the development of computational thinking, creativity, collaboration, and problem-solving skills. When integrated effectively, robotics does not function as an isolated technological activity but rather as a pedagogical tool that enriches subject learning across disciplines such as language, arts, history, science, and civic education.

The following best practices were identified through the **piloting of the Robots Meet Arts curriculum in several European educational contexts**. These practices highlight successful approaches for integrating robotics into everyday classroom learning while addressing important educational priorities such as inclusion, interdisciplinarity, and active student participation.

The practices demonstrate how robotics activities can be embedded in **curricular subjects**, structured to support diverse learners, and implemented using both digital and unplugged approaches. They also show how collaborative learning, reflection, and experimentation can help students understand computational concepts in meaningful and engaging ways.

Together, these best practices provide practical guidance for educators and schools seeking to integrate robotics into teaching and learning. They illustrate how robotics can support inclusive and interdisciplinary education while fostering the digital and creative competences needed in contemporary classrooms.

Co-funded by the European Union

Overview of Curriculum Best Practices

Best Practice Title	Country	Short Description
• Gradual Progression Model	Greece	• Robotics learning follows a structured progression from unplugged activities to block programming and robotics, allowing students to gradually develop computational thinking skills.
• Peer-Led Inclusive Robotics Learning	Greece	• Students with prior robotics experience support peers through mentoring roles, creating an inclusive environment where learners of different abilities collaborate.
• Visual and Multimodal Scaffolding for Inclusive Robotics Learning	Greece	• Visual supports, demonstrations, and hands-on activities help students with diverse linguistic and learning needs understand robotics concepts.
• Playful Approaches for Inclusive Engagement in Computational Thinking	France	• Playful activities such as games, creative challenges, and storytelling increase motivation and make robotics accessible to all learners.
• The Student Becomes the Robot: Embodying Robotics Without the Technical Object	France	• Students physically enact programming instructions, helping them understand sequences, algorithms, and logic through embodied learning.
• Multi-Platform Robotics Exploration to Broaden Computational Thinking	France	• Students explore multiple robotics tools and programming environments, understanding that computational thinking principles apply across technologies.
• Learning Boxes for Challenge-Based Robotics Learning	Spain	• Challenge-based “learning boxes” connect robotics with real-world problems, encouraging project-based learning, collaboration, and creative problem-solving.
• Structured Reflection After Robotics Activities	Spain	• Reflection sessions after activities help students analyse their learning process, identify challenges, and consolidate computational thinking concepts.
• Collaborative Debugging as a Classroom Learning Strategy	Spain	• Students work together to identify and fix programming errors, transforming mistakes into collective learning opportunities.
• Translating Everyday Tasks into Algorithms for Teaching Computational Thinking	Belgium	• Students break down familiar activities into precise step-by-step instructions, helping them understand the logic and precision required in programming.
• Learning Through Open Challenges	Belgium	• Teachers present open-ended challenges rather than detailed instructions, encouraging experimentation, autonomy, and problem-solving.
• Predict Before Programming	Belgium	• It encourages students to predict the outcome of a program before executing it on a robot or digital platform.

1. Gradual Progression Model

General Information

Country: Greece
Partner Organisation: Stimmuli For Social Change
Type of Best Practice: Curriculum Piloting

Description of the Practice

A key best practice identified during the piloting of the Robots Meet Arts Curriculum is the gradual progression model, where **students move from unplugged activities to block-based programming and finally to robotics kit implementation**. Teachers first introduced computational thinking concepts such as sequencing and algorithms through subject-based unplugged tasks, then reinforced them using Scratch or Scratch Jr., before applying them with robots like LEGO Spike or Vinci. This structured scaffolding reduced cognitive overload and ensured that technology enhanced learning rather than creating frustration.

Why is this a Best Practice?

This approach proved particularly effective because it addressed the **diverse readiness levels of students**. By gradually increasing complexity, teachers were able to support learners with limited digital experience while still challenging more advanced students.

Impact

The **progression model strengthened computational thinking skills** in a measurable and observable way. Students demonstrated improved sequencing accuracy, logical reasoning, and problem decomposition. Because the robotics implementation was the final stage of the learning process, it functioned as a motivational culmination rather than an overwhelming starting point. The tangible interaction with robots reinforced abstract concepts and increased persistence during problem-solving tasks. Students were more willing to debug and iterate because they understood the logic behind their actions.

Transferability

This best practice is highly transferable because it does not depend on a specific robot but on a pedagogical sequencing strategy. I**t can be implemented with different robotics kits or programming platforms and adapted across subjects and age groups**. The essential requirement is structured lesson planning that deliberately **scaffolds** computational thinking development before full robotics implementation.

2. Peer-Led Inclusive Robotics Learning

General Information

Country: Greece
Partner Organisation: Dimotiko Scholeio Plateos Imathias (DSPI)
Type of Best Practice: Curriculum Piloting

Description of the Practice

During the curriculum piloting, teachers implemented a **peer-tutoring model** in which **students with prior robotics experience supported classmates who faced greater barriers to participation,** including students with learning disabilities, autism, girls with limited exposure to robotics, and learners from diverse linguistic and cultural backgrounds. Experienced students were encouraged to act as mentors by explaining tasks, assisting classmates, and adapting their support to individual needs.

Why is this a Best Practice?

This approach addresses two key challenges in inclusive education: **engaging students who face barriers to participation while supporting the continued development of more advanced learners**. By positioning knowledgeable students as mentors, classroom diversity becomes a learning resource rather than an obstacle. Peer support created a low-pressure learning environment where students could progress at their own pace. At the same time, the tutoring role strengthened mentors' understanding of robotics concepts while developing communication, empathy, and leadership skills.

Impact

The practice increased **participation and confidence among students** who initially struggled to engage due to learning difficulties, language barriers, or lack of prior experience. Girls who were initially hesitant became more active participants through peer support. **Student tutors also benefited by developing leadership skills** and deepening their understanding of robotics concepts. Overall, collaboration improved, with students of different abilities and backgrounds working together towards shared learning goals.

Transferability

This model can be easily implemented in different educational contexts because it does not depend on a specific curriculum or type of robot. Teachers can apply the approach by identifying students with prior experience and encouraging them to support peers during robotics activities.

3. Visual and Multimodal Scaffolding for Inclusive Robotics Learning

General Information

Country: Greece
Partner Organisation: Dimotiko Scholeio Plateos Imathias (DSPI)
Type of Best Practice: Curriculum Piloting

Description of the Practice

During the curriculum piloting sessions, teachers used **visual and multimodal scaffolding strategies** to help students understand robotics and computational thinking concepts. Because the class included learners with different language levels, reading abilities, and special educational needs, teachers relied heavily on visual materials and demonstrations to support understanding. For example, **floor puzzle mats, visual command cards, and robot demonstrations were used to explain programming sequences and robot movements**. Teachers also modelled tasks step by step using the robots, allowing students to observe the programming process before attempting the activity themselves. In storytelling activities using Scratch and robotics tasks with Tale-Bot and LEGO Spike, visual storyboards and checklists helped students organise their ideas before moving to digital programming. These multimodal supports allowed students to **understand the logic of coding and sequencing even when linguistic explanations were difficult to follow**.

Why is this a Best Practice?

Visual and hands-on representations make abstract computational concepts more **accessible** for students with language barriers, lower reading levels, or special educational needs. By combining **visual materials, demonstrations, and physical interaction** with robots, teachers created a learning environment where students could understand programming concepts through observation and experimentation.

Impact

The use of visual scaffolding **increased student engagement and participation** across the group. Students who initially hesitated to speak or ask questions were able to follow the activities more confidently by relying on visual guidance and demonstrations. The approach also supported **collaboration**, as students could refer to shared visual materials when discussing robot commands or planning their tasks. As a result, learners with diverse abilities and language backgrounds were able to **participate actively** in robotics activities.

Transferability

This practice can **easily be transferred to other educational contexts** because it relies on simple strategies such as visual aids, demonstrations, and step-by-step modelling. Teachers can **adapt the approach using materials already available in classrooms**, including printed command cards, floor maps, storyboards, or diagrams.

4. Playful Approaches for Inclusive Engagement in Computational Thinking

General Information

Country: France
Partner Organisation: L.A.B (Laboratoire d'Aix-périmentation et de Bidouille)
Type of Best Practice: Curriculum Piloting

Description of the Practice

Across two piloting contexts — CERN (50 students aged 8–16) and a CLAS programme in La Rochelle (9 students aged 10–14, half with ADHD/ASD, from a priority education neighbourhood), computational thinking activities were designed **as games** before being framed as learning. Students passed balls to discover routing strategies, assembled physical blocks to program a classmate's dance, or navigated a grid through trial and error to build a reinforcement learning model. In every case, **the game mechanics carried the learning objectives**: students engaged because it was fun and learned because the game was well designed.

Why is this a Best Practice?

Play neutralised the barriers that typically exclude certain learners from computational thinking: no prior technical knowledge required, no screen anxiety, no reading-heavy instructions. Students with ADHD found focus through physical movement; beginners entered without intimidation; experienced students discovered genuine challenge beneath apparent simplicity. The game format created a level playing field across diverse profiles — age, background, prior experience, and specific learning needs became irrelevant to participation.

Impact

Both pilotings achieved 5/5 satisfaction for enjoyment and inclusion. At CERN, all 50 students completed activities regardless of prior knowledge. In La Rochelle, students with ADHD/ASD maintained engagement through multi-modal, movement-based activities. All students expressed desire to continue. Teachers unanimously rated activity suitability at 5/5 and all students expressed strong desire to continue with similar activities.

Transferability

Requires no specialised equipment — balls, coloured vests, printed cards, and a grid are sufficient. **The game-based entry point is language-independent and culturally neutral, making it directly replicable across European contexts**. It is especially valuable in inclusive settings (special needs, priority education, mixed-age groups) where traditional screen-based coding activities risk excluding certain learners.

5. The Student Becomes the Robot: Embodying Robotics Without the Technical Object

General Information

Country: France
Partner Organisation: L.A.B (Laboratoire d'Aix-périmentation et de Bidouille)
Type of Best Practice: Curriculum Piloting

Description of the Practice

During our piloting activities, we proposed activities where students **physically embody the role of the autonomous agent before touching any technical tool**. Learners experience from the inside what it means to execute instructions literally, explore an unknown environment, and build a model through trial, error, and reinforcement. This embodied approach **shifts the focus from the robot as an object to robotics as a way of thinking**: understanding how an agent perceives, decides, acts, and learns. The robot becomes a concept before it becomes a machine.

Why is this a Best Practice?

Too often, educational robotics is reduced to **making an expensive object move**. Students follow tutorials, press buttons, and watch a robot roll without understanding the fundamental questions underneath: how does an agent build knowledge? What makes an instruction precise enough to be executed? What is a model? By placing students in the agent's role, this practice **restores the intellectual core of robotics,** algorithmic reasoning, model construction, and autonomous decision-making, and proves these concepts can be explored **even if teachers cannot afford or access the technical material**. Robotics education becomes **accessible to any school, regardless of equipment budget**.

Impact

The approach produced a dual impact. Beginners discovered robotics concepts without intimidation or preconceptions. Experienced coders, conversely, were genuinely challenged: they realised that making a robot move is not the same as understanding how an autonomous agent learns and reasons.

Transferability

This practice is **zero-cost and immediately replicable** as it requires only physical space and basic materials. It works across all age groups (tested 8–16) and is particularly relevant for schools **where robotics equipment is unavailable or too expensive**. The principle i.e. teach what robotics is before introducing the technical object can be applied as a foundation to any Robots Meet Arts activity and adapted to any national curriculum.

6. Multi-Platform Robotics Exploration to Broaden Computational Thinking

General Information

Country: France
Partner Organisation: L.A.B (Laboratoire d'Aix-périmentation et de Bidouille)
Type of Best Practice: Curriculum Piloting

Description of the Practice

This practice introduces students to **multiple robotics platforms within a single learning sequence** to help them understand that computational thinking is independent of any specific technology. During the activity, students rotated between three robotics environments with very different programming approaches:

- **Cubetto**, a screen-free robot using physical programming blocks
- **Sphero Indi**, a robot controlled through colour-coded cards
- **Micro:bit**, programmed using block-based coding in MakeCode

Each group completed mission cards requiring them to apply the same logical model—moving a robot from a starting point to a target while avoiding obstacles—using the different systems. Through this rotation, students explored how the same computational concepts can be implemented through different programming interfaces.

Why is this a Best Practice?

This approach helps students understand that programming is based on general **computational principles rather than specific tools or platforms**. By encountering multiple robotics systems in a single activity, learners recognize common concepts such as sequencing, testing, and debugging across different technological environments.

Impact

Students with **different levels of prior experience were able to participate successfully**. Beginners discovered programming through accessible interfaces, while more experienced students deepened their understanding by translating the same logic into different systems. The approach also increased engagement, as students were curious to test each robot and compare how they worked. Exposure to different robotics tools helped students better understand the broader ecosystem of educational robotics.

Transferability

This practice can be implemented in many educational contexts by organising robotics activities as rotating workshops using different tools or programming environments. Schools can adapt the approach according to available equipment, combining screen-free robots, visual programming platforms, or microcontroller-based devices. The key principle is not the specific robots used but the intentional comparison between multiple systems, helping students develop transferable computational thinking skills.

Co-funded by the European Union

7. Learning Boxes for Challenge-Based Robotics Learning

General Information

Country: Spain
Partner Organisation: Maria Fortuny
Type of Best Practice: Curriculum Piloting

Description of the Practice

Learning boxes are inspired by challenge-based education and support hands-on, project-oriented learning. Each learning box is organised around a **real-world challenge** that **integrates robotics, coding, and multiple school subjects such as science, language, and social studies**. Students work collaboratively to explore the challenge, design solutions, and test their ideas using robotics tools. The activities follow a **structured learning sequence**: introduction of the challenge, exploration and research, practical experimentation with robotics, iterative improvement, and final presentation and reflection. Tasks within each learning box are scaffolded to gradually increase in complexity, allowing students to build confidence while developing computational thinking, creativity, and problem-solving skills.

Why is this a Best Practice?

Learning boxes provide a structured yet flexible framework for integrating robotics into meaningful learning experiences. By connecting robotics activities with real-world challenges, the approach increases student motivation and helps learners understand the relevance of computational thinking. The **challenge-based structure encourages collaboration, experimentation, and iterative problem solving**. Students learn not only technical skills but also **teamwork**, **communication**, and **reflective thinking** as they evaluate and improve their solutions.

Impact

Students showed **high levels of engagement and motivation** during the activities. The progressive structure helped learners develop **computational thinking** skills such as sequencing, debugging, and logical reasoning.

Transferability

Learning boxes can easily be adapted to different educational contexts because the structure is **flexible** and does **not depend on a specific robot or curriculum**. This approach can be implemented in a wide range of European classrooms using different robotics platforms and subject areas.

8. Structured Reflection After Robotics Activities

General Information

Country: Spain
Partner Organisation: Maria Fortuny
Type of Best Practice: Curriculum Piloting

Description of the Practice

This practice integrates structured reflection phases at the end of robotics-based learning activities to help students consolidate their understanding of both computational thinking and subject content. After completing programming or robotics tasks, students participate in **guided discussions** where they **analyze the strategies they used, the problems they encountered, and the solutions they developed**. Reflection activities may include short **group discussions, teacher-guided questions, peer feedback, or brief presentations of each group's work**. Students are encouraged to explain how they programmed the robot, what difficulties they faced during the process, and how they debugged errors.

Why is this a Best Practice?

Structured reflection strengthens learning by helping students become aware of their own thinking processes. When students describe how they solved programming challenges, they develop **metacognitive skills** and gain a clearer understanding of computational concepts such as sequencing, debugging, and algorithmic thinking. **Reflection also encourages collaborative learning**. Listening to other groups' solutions exposes students to different strategies and reinforces the idea that programming problems can be approached in multiple ways. In addition, **discussing difficulties openly helps normalize errors as part of the learning process**.

Impact

Students become more **confident** in identifying errors and proposing solutions during programming activities. Reflective discussions help consolidate both computational **thinking skills and subject knowledge,** as students connect their robotics work with the learning objectives of the lesson. The practice also strengthens **communication skills** and encourages more **active participation** during group activities.

Transferability

This practice can be easily implemented in any classroom using educational robotics. It does **not require additional equipment or resource**s, **only dedicated time at the end of activities for guided discussion and reflection.**

Co-funded by the European Union

9. Collaborative Debugging as a Classroom Learning Strategy

General Information

Country: Spain
Partner Organisation: Maria Fortuny
Type of Best Practice: Curriculum Piloting

Description of the Practice

This practice focuses on the use of collaborative debugging as a learning strategy during robotics and programming activities. When students encounter errors in their robot programs, the problem is not solved individually or immediately by the teacher. Instead, the issue becomes a shared challenge for the group or the entire class. Students are encouraged to observe the robot's behavior, identify where the sequence fails, and discuss possible solutions together. Groups explain their code, test alternative commands, and collectively analyze the results. The teacher facilitates the process by guiding the discussion and encouraging students to justify their reasoning.

Why is this a Best Practice?

Collaborative debugging helps students understand that mistakes are a natural and valuable part of computational thinking. Instead of focusing only on obtaining the correct result, students learn to analyze problems systematically and improve their solutions through iteration. The collaborative nature of debugging **encourages communication and peer learning. Students exchange ideas, explain their reasoning, and learn from different approaches to solving the same problem.** This process **strengthens** both **computational thinking skills** and **collaborative problem-solving abilities.** In addition, the strategy helps **reduce anxiety** related to programming errors. When mistakes are addressed collectively, **students feel more comfortable experimenting and testing new ideas**.

Impact

Students developed stronger **problem-solving** and **analytical skills** as they learned to identify and correct programming errors. The process improved their understanding of sequencing, logical reasoning, and algorithmic thinking. Teachers observed that collaborative debugging **increased classroom engagement and participation.** It also fostered a **supportive learning environment where errors were viewed as opportunities for improvement** rather than failures.

Transferability

Collaborative debugging can be implemented in any classroom using robotics or coding activities. It does not require additional resources, only a teaching approach that encourages discussion, experimentation, and peer support.

10. Translating Everyday Tasks into Algorithms for Teaching Computational Thinking

General Information

Country: Belgium
Partner Organisation: De Ark
Type of Best Practice: Curriculum Piloting

Description of the Practice

This practice introduces computational thinking by asking students to **translate familiar everyday activities into precise step-by-step instructions, similar to writing computer code**. Instead of starting directly with programming software, students first work with real-life processes that they already understand, such as preparing breakfast. Students write a sequence of instructions describing how to complete the task and then test the instructions by having a "human robot" follow them exactly as written. If the instructions are incomplete or ambiguous, the activity often leads to unexpected or humorous outcomes, helping students realize the importance of precision in programming.

Why is this a Best Practice?

This approach builds computational thinking using situations that are already familiar to students. Because learners have prior knowledge of everyday activities, they can focus on understanding the logic of algorithms rather than struggling with new technical tools.

Impact

Students develop key computational thinking skills, particularly decomposition, sequencing, and debugging. They become aware that programming requires precise instructions and systematic problem-solving. Teachers observed that students gained a clearer understanding of how programming works and became more confident in writing and testing algorithms.

Transferability

This practice is highly transferable because it requires no digital tools or specialized equipment. Teachers can easily apply the method using any familiar real-life task, such as preparing food, organizing a bag, or following a daily routine.

11. Learning Through Open Challenges

General Information

Country: Belgium
Partner Organisation: De Ark
Type of Best Practice: Curriculum Piloting

Description of the Practice

This practice promotes computational thinking by presenting students with **open-ended challenges rather than detailed step-by-step instructions**. Instead of following predefined procedures, learners are encouraged to explore possible solutions, test ideas, and discover programming logic through experimentation. In robotics and coding activities, students receive a clear goal or challenge—for example, programming a robot to complete a task or designing a simple interactive program. Teachers provide guidance and support when needed but intentionally avoid giving a complete solution. Students therefore work individually or in small groups to plan, test, and refine their programs.

Why is this a Best Practice?

Open challenges stimulate curiosity and engagement by transforming programming tasks into meaningful problems to solve rather than procedures to follow. This method reflects the authentic nature of programming, where solutions are rarely predetermined and often require experimentation and iteration.

Impact

Teachers observed that students became more independent and confident when working with robotics and programming tasks. The challenge-based approach increased motivation and encouraged learners to persist when facing difficulties. Students developed stronger problem-solving abilities and gained a deeper understanding of programming concepts because they actively constructed their own solutions rather than simply reproducing instructions.

Transferability

This practice can be easily applied in different educational contexts and does not depend on specific robotics platforms or digital tools. Teachers can introduce open challenges in a wide range of subjects and activities involving robotics, programming, or computational thinking. Because the method focuses on inquiry and exploration, it can be adapted to different age groups and classroom contexts, making it highly transferable across European schools.

12. Predict Before Programming

General Information

Country: Belgium
Partner Organisation: De Ark
Type of Best Practice: Curriculum Piloting

Description of the Practice

This practice encourages students to predict the outcome of a program before executing it on a robot or digital platform. Before pressing the “run” or “start” command, students are asked to observe the sequence of instructions and discuss what they expect the robot to do. Teachers may invite students to explain their reasoning individually or within small groups. Students analyse the instructions step by step, imagining how the robot will move or behave according to the programmed sequence. Only after this prediction phase do they execute the program and observe the actual result. This short reflective step transforms programming from a trial-and-error activity into a process of reasoning and hypothesis testing.

Why is this a Best Practice?

Encouraging students to predict the outcome of a program strengthens computational thinking skills, particularly sequencing, logical reasoning, and algorithmic understanding. By analysing instructions before execution, learners begin to understand how each command influences the robot’s behaviour. The practice also reduces random experimentation. Instead of repeatedly testing code without reflection, students develop a habit of planning and anticipating results. When predictions differ from the robot’s actual behaviour, students naturally engage in debugging and discussion, deepening their understanding of the program. This approach is simple to implement yet highly effective in helping students connect abstract instructions with concrete robotic actions.

Impact

Students become more thoughtful and systematic when programming robots. The prediction phase encourages them to read and analyse their code carefully, improving both accuracy and problem-solving skills.

Transferability

The practice can be easily implemented in any robotics or coding activity regardless of the specific platform used. Teachers only need to introduce a brief “prediction moment” before running a program, making it a low-effort strategy with significant pedagogical benefits.

6. Recommendations for Future Implementation in EU Countries

Based on the experiences gathered through the teacher training programme and curriculum piloting activities, the Robots Meet Arts project identifies several recommendations for supporting the wider adoption of educational robotics across European education systems.

- **Strengthen teacher training in educational robotics.**
  Teacher confidence and pedagogical understanding are essential for the successful implementation of robotics activities in schools. Training programmes should therefore focus not only on the technical use of robotics tools but also on pedagogical strategies for integrating robotics into different subjects and supporting inclusive learning environments.

- **Teacher confidence and pedagogical understanding are essential for the successful implementation of robotics activities in schools.**
  Training programmes should therefore focus not only on the technical use of robotics tools but also on pedagogical strategies for integrating robotics into different subjects and supporting inclusive learning environments.

- **Integrate robotics across the curriculum.**
  Educational robotics should not be limited to STEM subjects. Instead, it can support interdisciplinary learning by connecting coding and computational thinking with subjects such as language learning, arts, history, civic education, and social studies. This interdisciplinary approach increases student motivation and demonstrates the broader relevance of digital competences.

- **Promote inclusive learning approaches.**
  Robotics activities should be designed to support students with diverse learning needs, linguistic backgrounds, and levels of digital confidence. Inclusive strategies may include visual scaffolding, hands-on activities, peer collaboration, and differentiated learning tasks.

- **Combine digital and unplugged learning activities.**
  Unplugged activities that explore computational thinking concepts without technology can provide accessible entry points for beginners and help reduce technical barriers. These activities can also be particularly valuable in schools with limited access to robotics equipment.

Encourage collaborative and problem-based learning.
Many successful robotics activities are based on collaboration, experimentation, and problem-solving. Group work, challenge-based tasks, and collaborative debugging encourage students to share ideas, learn from mistakes, and develop resilience in the learning process.

Support experimentation and creativity.
Robotics education should allow students to explore, test ideas, and develop creative solutions rather than simply following predefined instructions. Open-ended challenges and project-based learning activities help students develop deeper understanding and ownership of their learning.

Ensure accessibility and adaptability of robotics activities.
Robotics learning activities should be adaptable to different school contexts, available technologies, and national curricula. Practices that focus on general computational thinking concepts rather than specific tools are more easily transferable across educational systems.

Encourage collaboration between educational institutions and innovation centres.
Partnerships between schools, universities, and research institutions—such as the collaboration with CERN in the project—can provide valuable opportunities for innovation, experimentation, and the dissemination of good practices in robotics education.

7. Conclusions

The Robots Meet Arts project demonstrates that educational robotics can play a transformative role in primary education when implemented through inclusive, interdisciplinary, and pedagogically grounded approaches.

Through the development and piloting of a teacher training programme and a curriculum integrating robotics with arts and humanities subjects, the project explored how robotics can support both digital competence development and broader educational goals such as creativity, collaboration, and critical thinking.

The implementation activities carried out across multiple European contexts provided valuable insights into how robotics activities function in real classroom environments with diverse learners and varying levels of technological familiarity. These experiences highlighted the importance of teacher training, progressive learning pathways, and inclusive pedagogical strategies.

The best practices identified in this report illustrate that robotics education can be accessible and meaningful even in contexts with limited technological resources. By combining digital tools with unplugged activities, collaborative learning structures, and reflective practices, robotics can become an effective medium for exploring complex ideas and fostering active student engagement.

Ultimately, the project shows that robotics should not be viewed solely as a technological skill but as a pedagogical approach that supports interdisciplinary learning and inclusive education. By adopting the practices and recommendations presented in this report, educators and policymakers across Europe can contribute to the development of innovative learning environments that prepare students for the challenges of a rapidly evolving digital society.